

## GENERAL INFORMATION

This manual provides instruction in the following repairs:

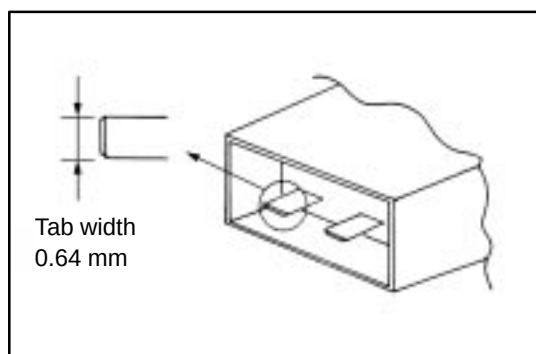
- How to Inspect for System Inspection
- Circuit Protection
- Terminal and Connector Repair Procedure

### NOTICE:

When inspecting or repairing the SRS AIRBAG, be sure to carefully read the precautionary instructions and procedure in the Repair Manual for the applicable model.

**A**

After any electrical repair is made, always test the circuit by operating the devices in the circuit. This confirms not only that the repair is correct, but also that the cause of the complaint was correctly identified.



### Terminal type number name

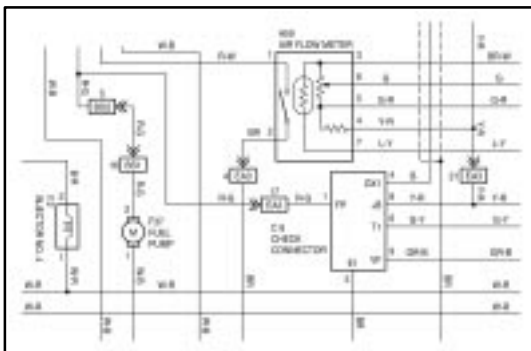
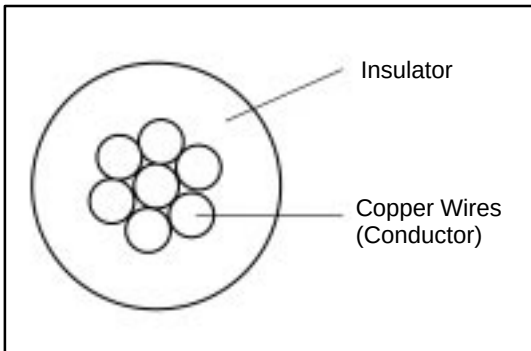
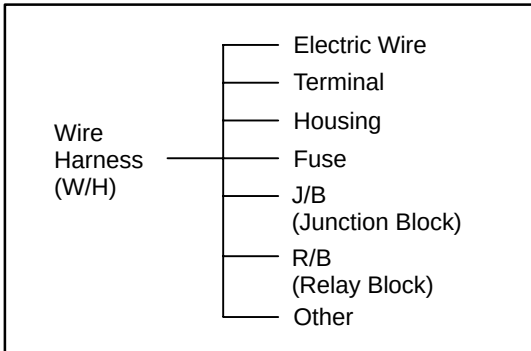
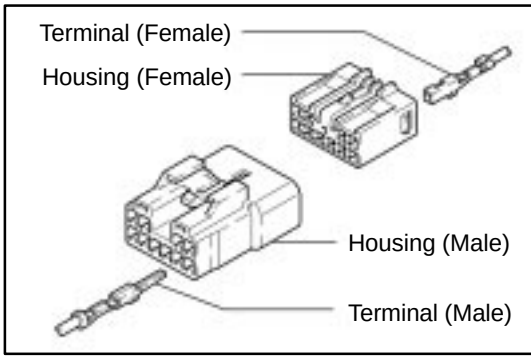
The terminal type number naming system has changed: the metric system will be used in place of the inch system. To be more specific, male tab width "in millimeters" will be used as terminal type number from now on — in place of male tab width in inches which has been in use so far.

Below is a table of comparison:

| Old terminal type No.<br>(inch) |   | New terminal type No.<br>(mm) |
|---------------------------------|---|-------------------------------|
| 025                             | → | 0.64                          |
| 040 (II, III, IV)               | → | 1.0 (II, III, IV)             |
| 050                             | → | 1.3                           |
| 070 (II)                        | → | 1.8 (II)                      |
| 090 (II)                        | → | 2.3 (II)                      |
| 187                             | → | 4.8                           |
| 250 (II)                        | → | 6.3 (II)                      |
| 305                             | → | 7.7                           |
| 312                             | → | 8.0                           |
| 375                             | → | 9.5                           |

For those connectors which are not shown above, the terminal type numbers remain unchanged.

## INTRODUCTION-GENERAL INFORMATION



|            |                  |
|------------|------------------|
| B = Black  | W = White        |
| L = Blue   | BR = Brown       |
| V = Violet | SB = Sky Blue    |
| R = Red    | G = Green        |
| P = Pink   | LG = Light Green |
| Y = Yellow | GR = Gray        |
| O = Orange |                  |

### What is Wire Harness?

The wire harness (W/H) is systems of electric wires for automobiles to connect all the electronics parts in automobile electrically and work them. As more electronics parts are introduced in automobiles recently, the electric wires used for the wire harness are increasing in number and the structure is becoming more complicated. As a result of that, there are hundreds of connectors, which is the parts which connects wires mechanically and electrically, in one vehicle. Therefore, numbers of the connector terminal (Terminal) or the connector housing (Housing) are designed to meet many kinds of uses of circuits. Various components have been improved to ensure the product reliability or realize a wider space in the vehicles.

### Wire Harness Components

Wire harness mainly consists of wires, terminals, or housings.

There are various components are designed for many parts of vehicles, such as one with high heat-resistance, water-resistance, or bending ability, ones have different current capacities, or ones are hardly influenced by electromagnetic noise.

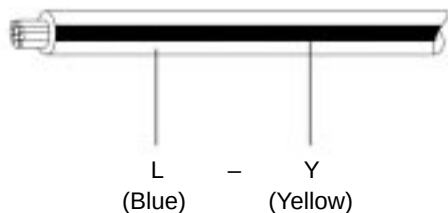
### Electric Wire

The electric wires used for the wire harness consists of the conductor made from numbers of twisted mild copper wire with a diameter of less than 0.5 mm and the insulator surrounding the conductor.

The insulator is generally made from vinyl chloride and covers the conductor with even thickness. The insulators are color-coded in order to distinguish each wire. The base colors or the stripe patterns is used to make difference between insulators. Each color of wires is indicated by the abbreviation in the repair manual and the electrical wiring diagram.

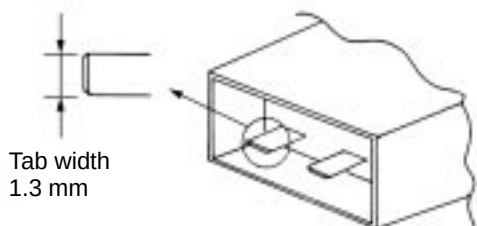
## INTRODUCTION-GENERAL INFORMATION

Example: L-Y



The first letter indicates the basic wire color and the second letter indicates the color of the stripe.

A

**Terminal and housing**

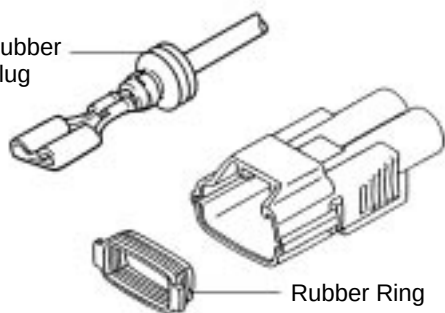
Terminal connects wires and housing insulates connecting parts.

There are the male terminal and the female terminal. The types of terminals are decided by tab width of male terminal. And the terminal with the rubber plug or the rubber ring is used in the part, such as the engine compartment, which become wet. For the circuit with slight current at EFI system or ABS system, the gold-plated terminal is introduced for ensuring reliabilities.

As the number of the circuit is increasing recently, there are new types of parts introduced. For example, there is the hybrid type housing, which is a combination of terminals with different tab width, such as the power source terminal or the signal terminal. Also, new type of connector such as the double lock housing, which is designed with the retainer in addition to the lance to prevent terminal from slipping off, is available. The new type of connectors are produced to realize higher product reliabilities and utilized widely recently. The major characteristics of these new connectors are shown in the table 1.

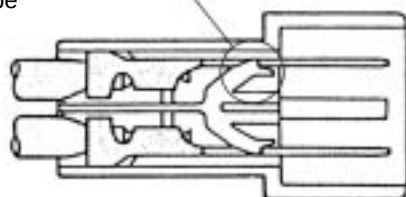
There are two types of lances: housing lance, which is inside the housing and terminal lance, which is inside the terminal.

Rubber Plug

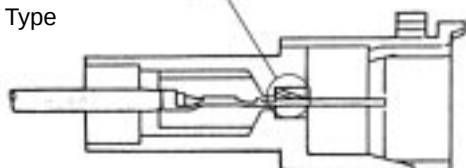


Rubber Ring

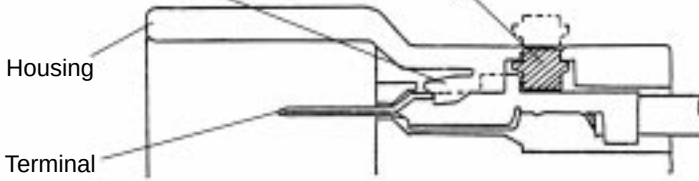
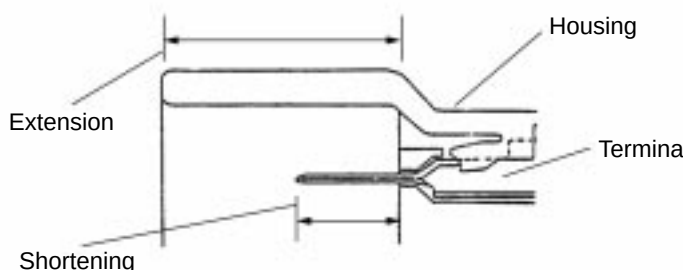
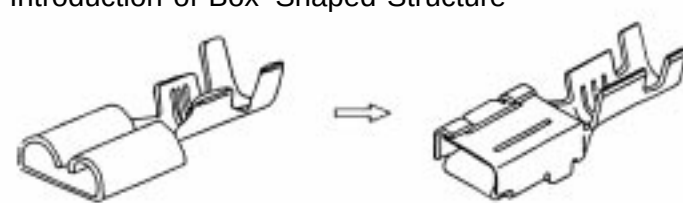
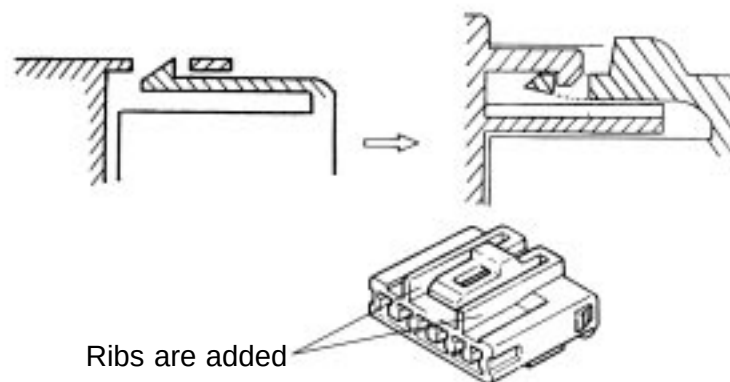
Housing Lance Type



Terminal Lance Type



**Table 1 : Characteristics of the new types of connector (Improvement)**

| Characteristics (Improvement)                                                                                                                                      | Note                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Double Lock</p> <p>Lance (Primary Lock)      Retainer (Double Lock)</p>     | <p>If terminal is not inserted to housing correctly, the retainer does not fit.</p>                                                 |
| <p>2. Extension of Housing</p>                                                   | <p>This is to prevent deformation of the terminal when it is inserted diagonally.</p>                                               |
| <p>3. Change of Contact Structure and Introduction of Box-Shaped Structure</p>  | <p>Expansion of contact section<br/>Stabilization of contact pressure</p>                                                           |
| <p>4. Change of Locking Shape</p>  <p>Ribs are added</p>                        | <p>This improves the close fit of locking and you can hear the click sound and feel that the connector is completely installed.</p> |

## HOW TO PERFORM FOR SYSTEM INSPECTION

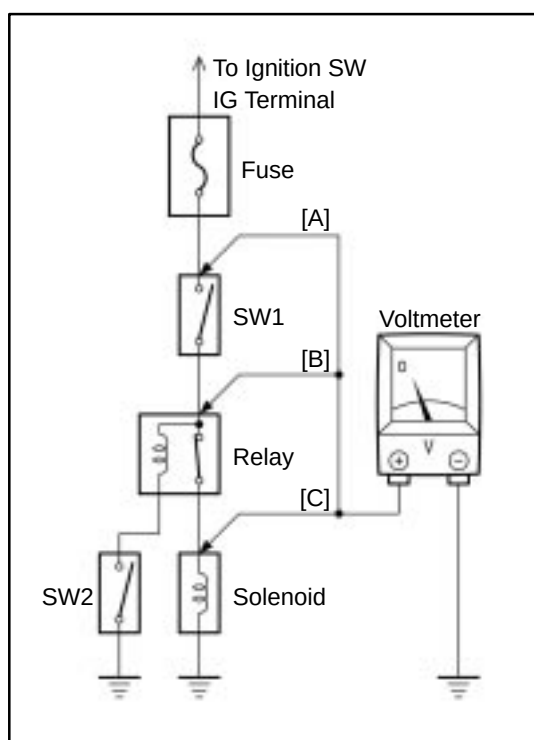
This inspection procedure is a simple troubleshooting which should be carried out on the vehicle during system operation and is based on the assumption of system component trouble

Always inspect the trouble taking the following items into consideration:

- Ground point fault
- Open or short circuit of the wire harness
- Connector or terminal connection fault
- Fuse or fusible link fault

### NOTICE:

- This is an on-vehicle inspection during system operation. Therefore, inspect the trouble with due regard for safety.
- If connecting the battery directly, be careful not to cause a short circuit, and select the applicable voltage.



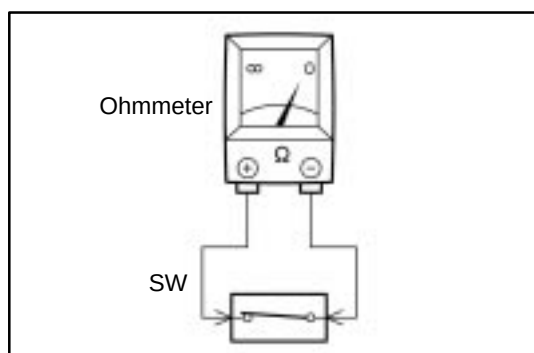
### 1. Voltage Check

- Establish conditions in which voltage is present at the check point.

Example:

- [A] – Ignition SW on
- [B] – Ignition SW and SW 1 on
- [C] – Ignition SW, SW 1 and Relay on (SW 2 off)

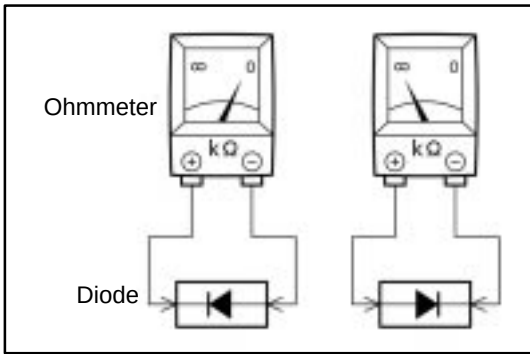
- Using a voltmeter, connect the negative (–) lead to a good ground point or negative (–) battery terminal and the positive (+) lead to the connector or component terminal. This check can be done with a test bulb instead of a voltmeter.



### 2. Continuity and Resistance Check

- Disconnect the battery terminal or wire so there is no voltage between the check points.
- Contact the two leads of an ohmmeter to each of the check points.

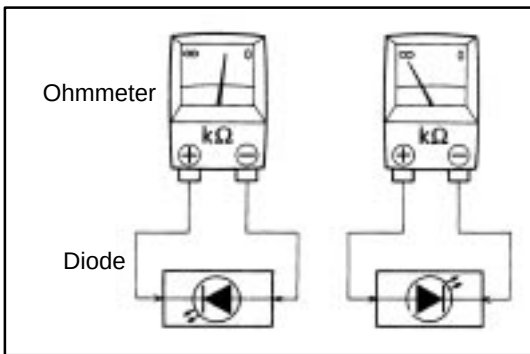
## INTRODUCTION-HOW TO PERFORM FOR SYSTEM INSPECTION



If the circuit has diodes, reverse the two leads and check again.

When touching the negative (–) lead to the diode positive (+) side and the positive (+) lead to the negative (–) side, there should be continuity. When touching the two leads in reverse, there should be no continuity.

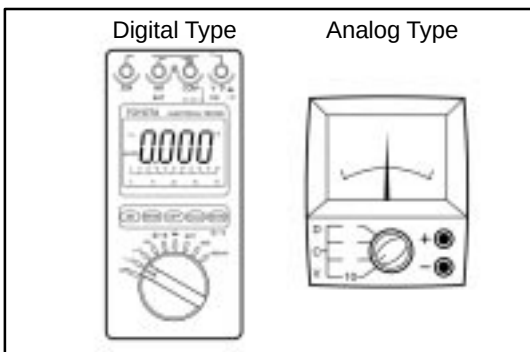
**HINT:** Specifications may vary depending on the type of tester, so refer to the tester's instruction manual before performing the inspection.



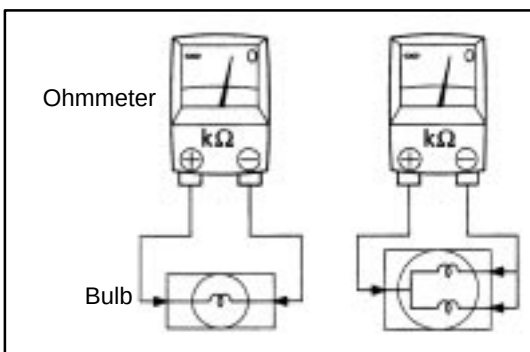
Check LED (Light Emitting Diode) in the same manner as that for diodes.

**HINT:**

- Use a tester with a power source of 3V or greater to overcome the circuit resistance.
- If a suitable tester is not available, apply battery voltage and check that the LED lights up.

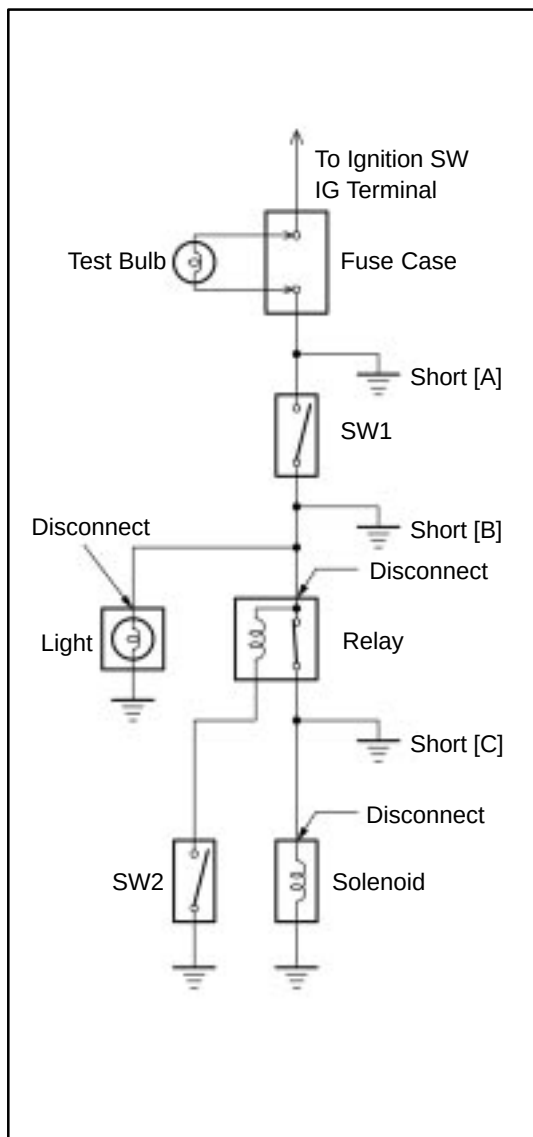


- (c) Use a volt/ohmmeter with high impedance (10kΩ/V minimum) for troubleshooting of the electrical circuit.



### 3. Bulb Check

- Remove the bulb.
- There should be continuity between the respective terminals of the bulb together with a certain amount of resistance.
- Apply the two leads of the ohmmeter to each of the terminals.
- Apply battery voltage and check that the bulb light up.



#### 4. Finding a Short Circuit

- Remove the blown fuse and eliminate all loads from the fuse.
- Connect a test bulb in place of the fuse.
- Establish conditions in which the test bulb comes on.

Example:

- [A] – Ignition SW on  
 [B] – Ignition SW and SW 1 on  
 [C] – Ignition SW, SW 1 and Relay on (Connect the Relay) and SW 2 off (or disconnect SW 2)

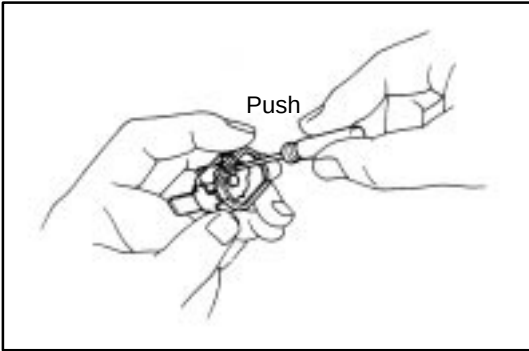
- Disconnect and reconnect the connectors while watching the test bulb. The short lies between the connector where the test bulb stays lit and the connector where the bulb goes out.
- Find the exact location of the short by lightly shaking the problem wire along the body.

#### CAUTION:

- Do not open the cover or the case of the ECU unless absolutely necessary. (If the IC terminals are touched, the IC may be destroyed by static electricity.)**
- When replacing the internal mechanism (ECU part) of the digital meter, be careful that no part of your body or clothing comes in contact with the terminals of leads from the IC, etc. of the replacement part (spare part).**

## CIRCUIT PROTECTION

All electrical circuits are protected against excessive loads which might occur because of shorts or overloads in the wiring system. Such protection is provided by a fuse, circuit breaker, or fusible link. A short circuit may cause a fuse to blow or a circuit breaker to open.



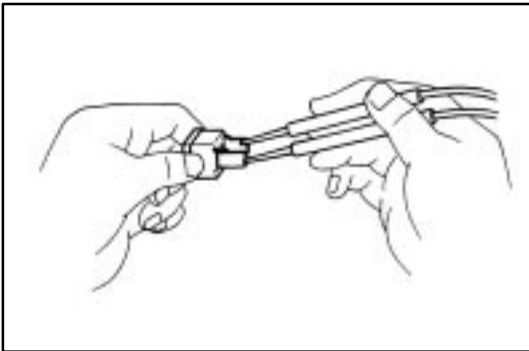
### RESET CIRCUIT BREAKER

#### 1. Remove Circuit Breaker

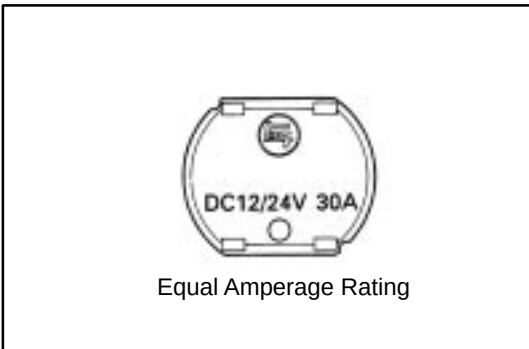
- (a) Disconnect the negative (–) cable from the battery.
- (b) Remove the circuit breaker.

#### 2. Reset Circuit Breaker

- (a) Insert the needle into the reset hole and push it.



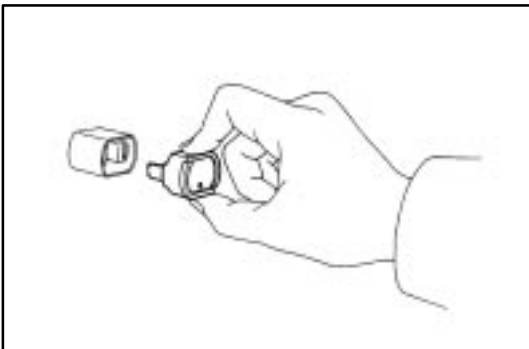
- (b) Using an ohmmeter, check that there is continuity between both terminals of the circuit breaker. If continuity is not as specified, replace the circuit breaker.



**HINT:** If replacing the circuit breaker, be sure to replace it with a breaker with an equal amperage rating.

#### 3. Install Circuit Breaker

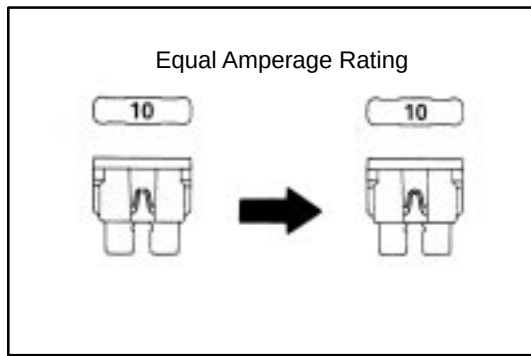
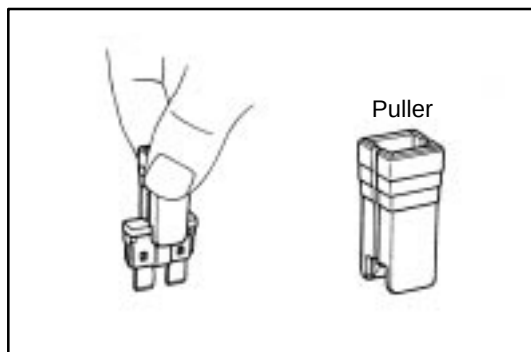
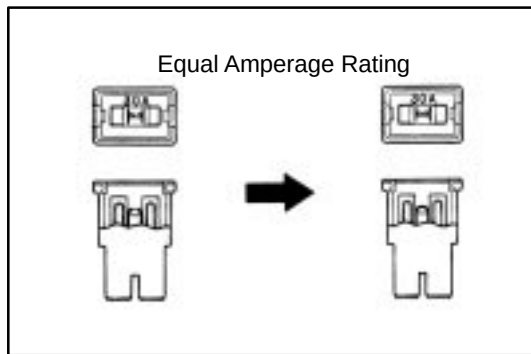
- (a) Install the circuit breaker.
  - (b) Connect the negative (–) cable to the battery.
- HINT:** If a circuit breaker continues to cut out, a short circuit is indicated. Have the system checked by a qualified technician.





**REPLACEMENT OF FUSE AND FUSIBLE LINK**

**HINT:** If replacing the fuse or fusible link, be sure to replace it with a fuse or fusible link with an equal amperage rating.

**A****NOTICE:**

1. **Turn off all electrical components and the ignition switch before replacing a fuse or fusible link. Do not exceed the fuse or fusible link amperage rating.**
2. **Always use a fuse puller for removing and inserting a fuse. Remove and insert straight in and out without twisting. Twisting could force open the terminals too much, resulting in a bad connection.**

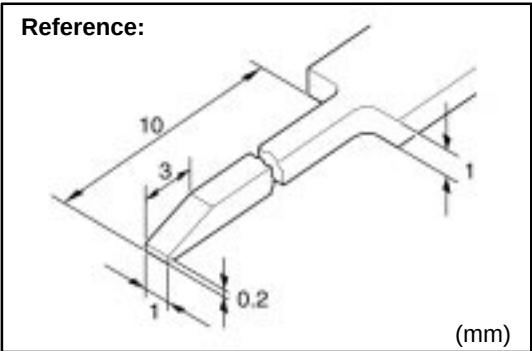
If a fuse or fusible link continues to blow, a short circuit is indicated. The system must be checked by a qualified technician.

**HINT:** The puller is located at Junction Block No.2.

PREPARATORY ITEMS

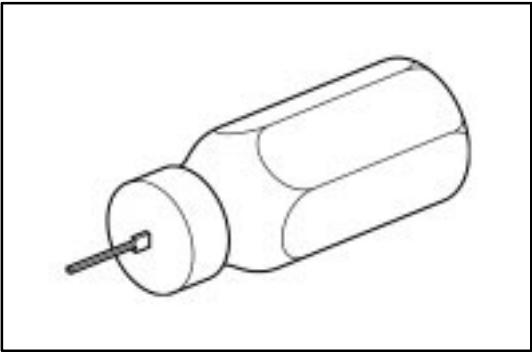
|        |               |                                                                 |                                       |
|--------|---------------|-----------------------------------------------------------------|---------------------------------------|
| Tool   | Crimping tool | AMP Part No. 169060                                             | For caulking press sleeves            |
|        | Special tool  | Refer to the following illustration                             | To remove the terminal                |
|        | SST           | 09991-00500                                                     | To remove the 0.64 connector terminal |
| Gauge  | Caliper       | 0-150 mm                                                        | To measure the diameter of the core   |
| Others | Press sleeve  | 82999-12010 (Red)<br>82999-12020 (Blue)<br>82999-12030 (Yellow) | To connect wires                      |
|        | Silicon tape  | 08231-00045                                                     | To waterproof the connected section   |

B



Special Tool

HINT: To remove the terminal from the connector, please construct and use the special tool or like object shown on the left.



Preparation of the SST to release the 0.64 connector terminal.  
SST 09991-00500

HINT:  
This is a SST for releasing terminals from 0.64 connectors.

## CONNECTOR REPAIR

The repair parts now in supply are limited to those connectors having common shapes and terminal cavity numbers. Therefore, when there is no available replacement connector of the same shape or terminal cavity number, please use one of the alternative methods described below. Make sure that the terminals are placed in the original order in the connector cavities, if possible, to aid in future diagnosis.

1. When a connector with a different number of terminals than the original part is used, select a connector having more terminal cavities than required, and replace both the male and female connector parts.

EXAMPLE: You need a connector with six terminals, but the only replacement available is a connector with eight terminal cavities. Replace both the male and female connector parts with the eight terminal part, transferring the terminals from the old connectors to the new connectors.

2. When several different type terminals are used in one connector, select an appropriate male and female connector part for each terminal type used, and replace both male and female connector parts.

EXAMPLE: You need to replace a connector that has two different types of terminals in one connector. Replace the original connector with two new connectors, one connector for one type of terminal, another connector for the other type of terminal.

3. When a different shape of connector is used, first select from available parts a connector with the appropriate number of terminal cavities, and one that uses terminals of the same size as, or larger than, the terminal size in the vehicle. The wire lead on the replacement terminal must also be the same size as, or larger than, the nominal size of the wire in the vehicle. ("Nominal" size may be found by looking at the illustrations in the section F or by direct measurement across the diameter of the insulation). Replace all existing terminals with the new terminals, then insert the terminals into the new connector.

EXAMPLE: You need to replace a connector that is round and has six terminal cavities. The only round replacement connector has three terminal cavities. You would select a replacement connector that has six or more terminal cavities and is not round, then select terminals that will fit the new connector. Replace the existing terminals, then insert them into the new connector and join the connector together.

## TERMINAL REPLACEMENT

These steps **must** be followed when replacing a terminal.

**Step 1. Identify the connector and the terminal type.**



**Step 2. Disconnect the terminal from the connector.**



**Step 3. Replace the terminal.**



**Step 4. Install the terminal into the connector.**

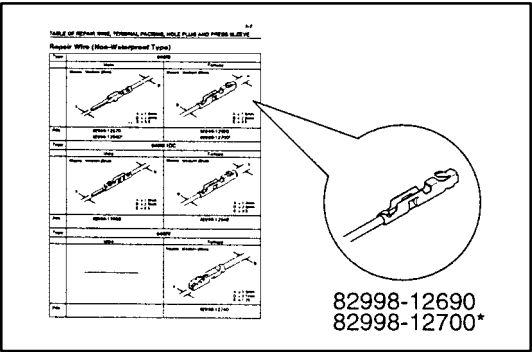
**B**

Step 1. Identify the connector and the terminal type.

Confirm whether the connector you require is the non-waterproof, waterproof or combined terminal type connector from the pictures provided in the following charts.

| Connector              | Description                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------|
| Non-Waterproof Type    | Those connectors which are not of the waterproof or combined terminal type.                     |
| Waterproof Type        | Waterproof material (hole plug or terminal packing) is provided on the terminal/connector body. |
| Combined Terminal Type | Terminals of different shape/size are located in one connector.                                 |

B



1. Terminal with Wire

Identify the part number of the terminal with wire by picking out the corresponding illustration from the pictures provided on the following charts.

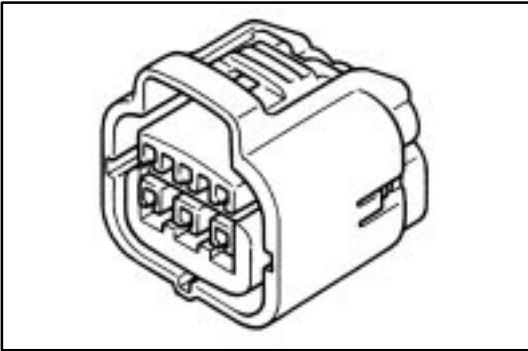
HINT: In some of the illustrations, there are two part numbers for the same illustration of terminal with wire, because although the shapes of the terminals are completely the same, there is a difference in with /without gold-plating (gilded).

Remark: The length of the wire connected to the terminal is approximately 150 mm.

## TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT

### NOTICE:

- When the terminal used is gold plated, be sure to replace it with a gold plated terminal when making repairs.
- Do not use male and female terminals which are made of different materials from each other.



## 2. Connector Body

- Identify the part number of the corresponding terminal with wire according to the above-mentioned 1.
- Identify the part number of the connector body by picking out the correct one from the illustrations according to the number of terminals and the shape.

B

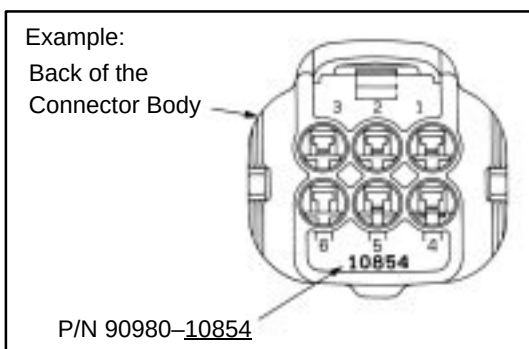
### HINT:

- If you can find no connector which matches the type you require, pick out male and female connector bodies as a set which have more terminals than you require.

For the combined terminal type connector body, pick out and appropriate connector body for each terminal being used.

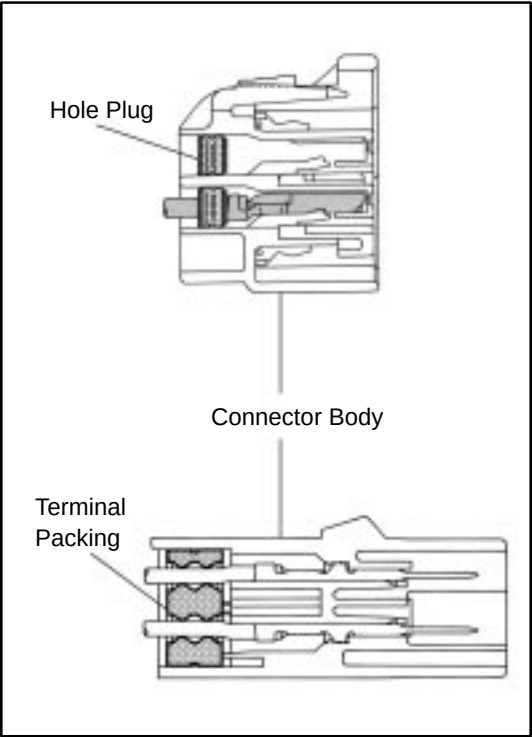
### Example:

When two types of terminal are used in one connector body, pick out a male and female for two different connector bodies.



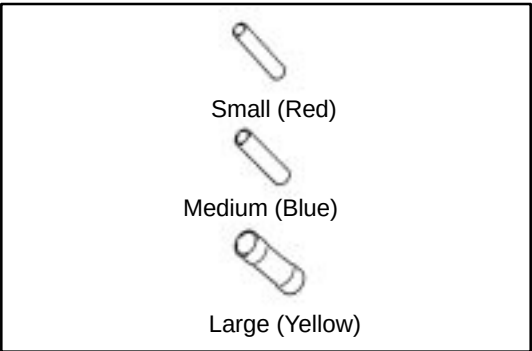
- The part number has been stamped on the back of the connector body. (This will continue to be done to new types of connector.)

B



### 3. Hole Plug/Terminal Packing

- (a) Identify the part number of the appropriate terminal with wire/connector body from the illustrations of waterproof type connector according to the above-mentioned 1 and 2.
- (b) Identify the part number of the hole plug/terminal packing from the corresponding illustration.

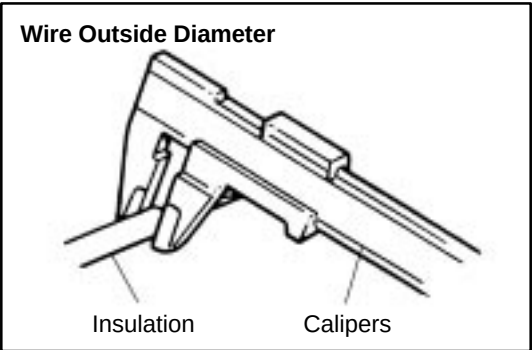


### 4. Sleeve

When connecting two wires using a sleeve, select the sleeve according to the following table.

| Wire Outside Diameter | Sleeve Size (Color) |
|-----------------------|---------------------|
| 0.2–1.0 mm            | Small (Red)         |
| 1.0–2.0 mm            | Medium (Blue)       |
| 3.0–5.0 mm            | Large (Yellow)      |

- (a) When size is based on the outside diameter of the wire.



# TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT

| * \ * | 0.3 | 0.5 | 0.85 | 1.25 | 2 | 3 | 5 |
|-------|-----|-----|------|------|---|---|---|
| 0.3   | S   | M   | M    | M    | M | L | L |
| 0.5   | M   | M   | M    | M    | M | L | L |
| 0.85  | M   | M   | M    | M    | L | L | L |
| 1.25  | M   | M   | M    | M    | L | L | L |
| 2     | M   | M   | L    | L    | L | L | — |
| 3     | L   | L   | L    | L    | L | L | — |
| 5     | L   | L   | L    | L    | — | — | — |

\* Nominal size of the wire

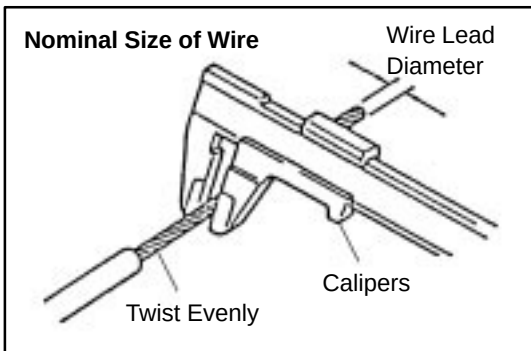
(b) When size is based on the nominal size of the wire.

## Sleeve size

S : Small (Red)

M : Medium (Blue)

L : Large (Yellow)

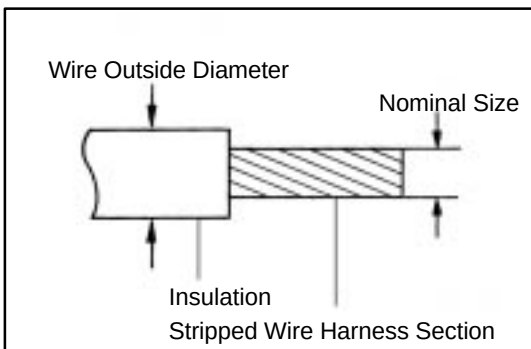


HINT: To calculate the "Nominal Size" of the wire.

Nominal size =

$$\frac{3.14 \times (\text{Diameter of stripped wire harness})^2}{4}$$

B



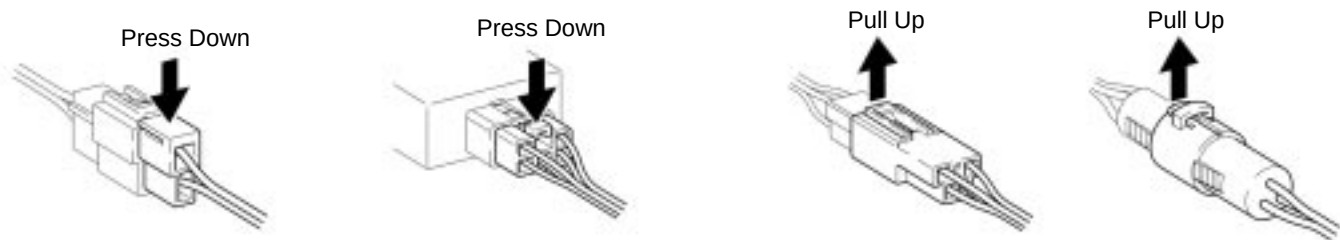
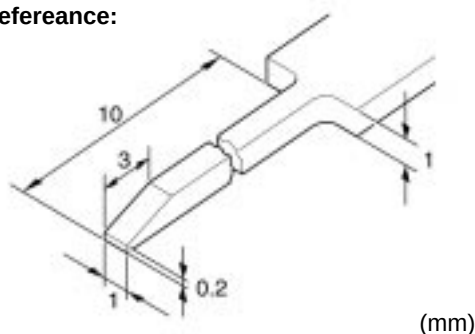
HINT: Outside Diameter and Nominal Size



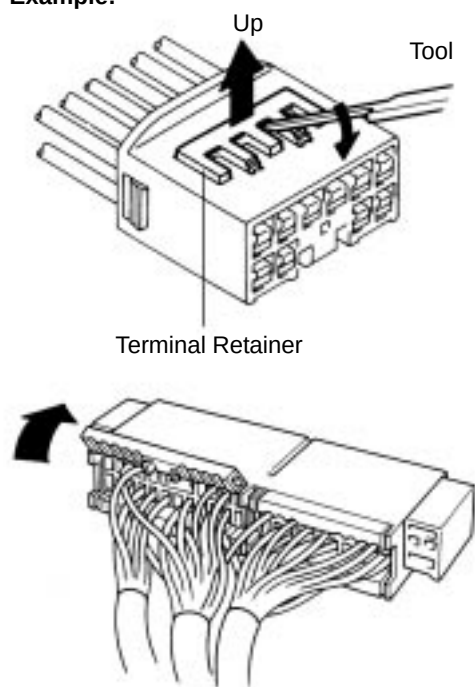
**Step 2. Disconnect the Terminal from the Connector.****1. Disconnect Connector**

To pull apart the connectors, pull on the connector itself, not the wire harness.

HINT: Check to see what kind of connector you are disconnecting before pulling apart.  
Press down type is mainly used.

**Example:****Reference:****2. Prepare the Special Tool**

HINT: To remove the terminal from the connector, please construct and use the special tool or like object shown on the left.

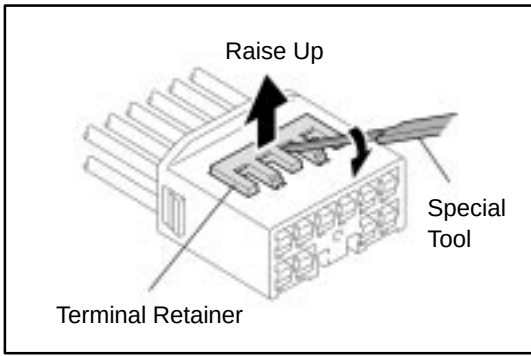
**Example:****3. Disengage the double locking device or terminal retainer.**

- (a) Locking device must be disengaged before the terminal locking clip can be released and the terminal removed from the connector.
- (b) Use a special tool or the terminal pick to unlock the double locking device.

**NOTICE:**

**Do not remove the terminal retainer from connector body.**

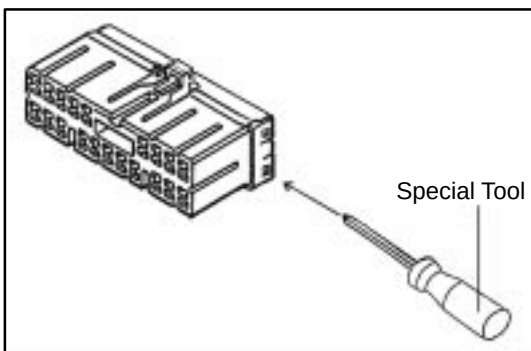
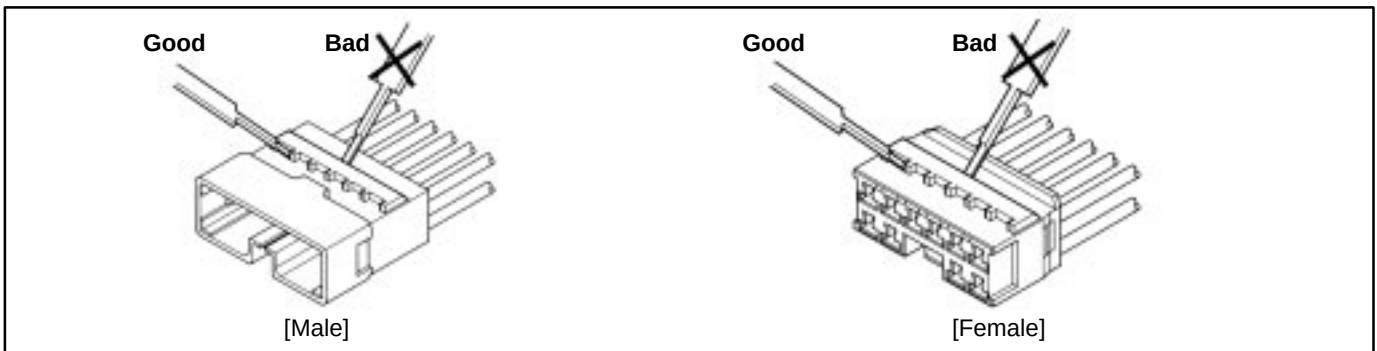
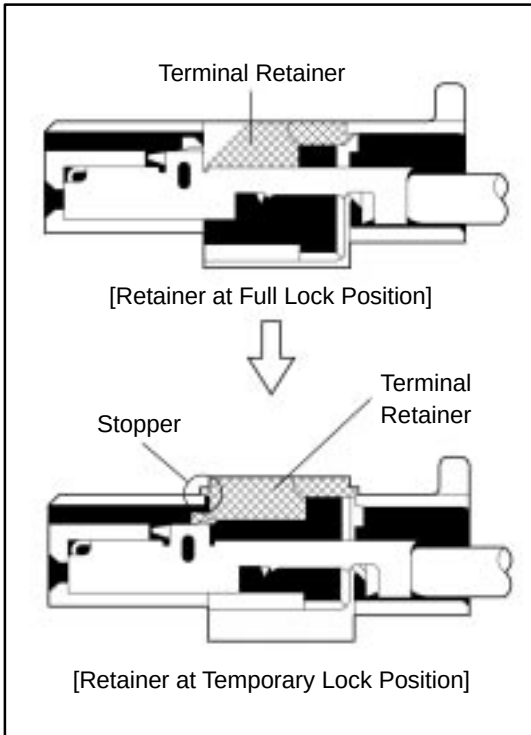
# TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT



Type A (For 1.0II, 1.8, 2.3II, 4.8 and 8.0 of Non-Waterproof Type Connector)

- (1) Using the special tool, raise the retainer up to the temporary lock position.

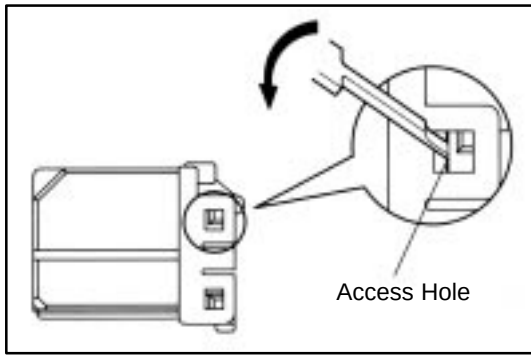
HINT: The needle insertion position varies according to the connector's shape (number of terminals, etc.), so check the position before inserting it.



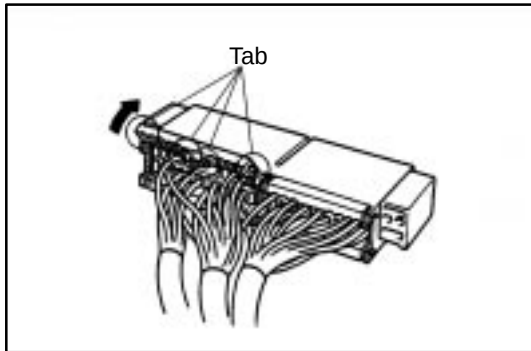
Type B (For 1.8, 1.0 and TLC of Non-Waterproof Type Connector)

B

## TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT



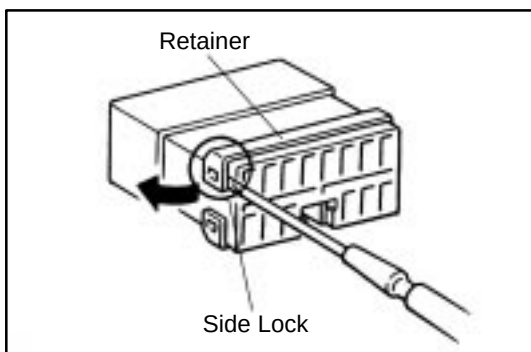
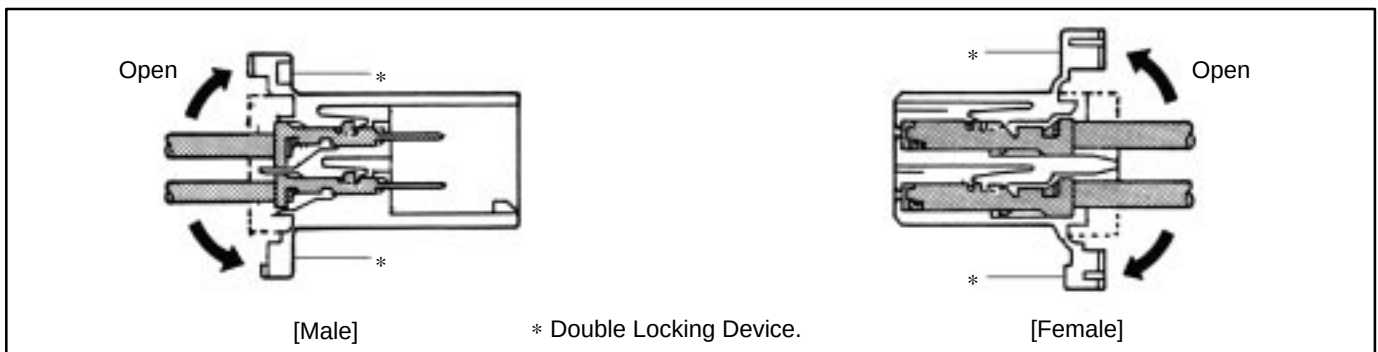
- (1) Press the special tool at a 45° angle into the locking lug access hole as shown.  
Raise the double locking device up as far as possible.



- (2) Remove the special tool and open the double locking device

**NOTICE:**

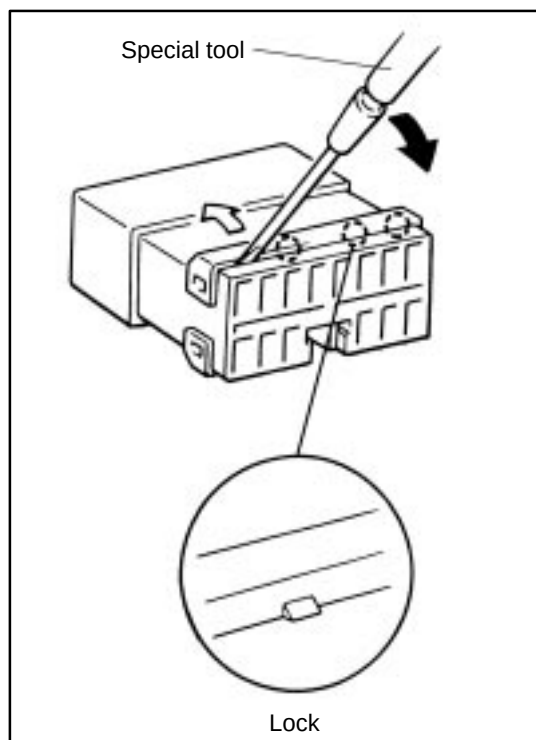
**TLC housing is not be reusable. Please replace it with the new one after replacement of the terminal.**



**Type C (For TNS, FTC Type Connector)**

- (1) Widen the side lock part of the retainer from side to side using the special tool.

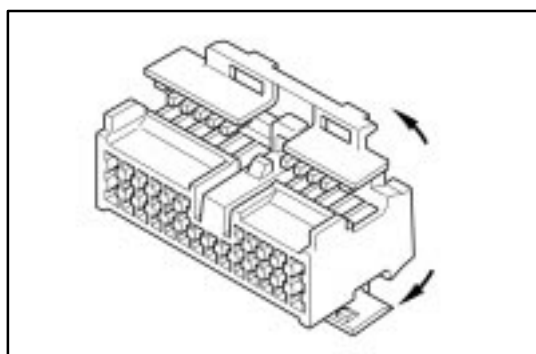
# TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT



- (2) Inset the special tool into the chink between the retainer and the terminal itself. Then pry it to the direction of the arrow shown in the illustration and push the retainer up to release the lock.

## NOTICE:

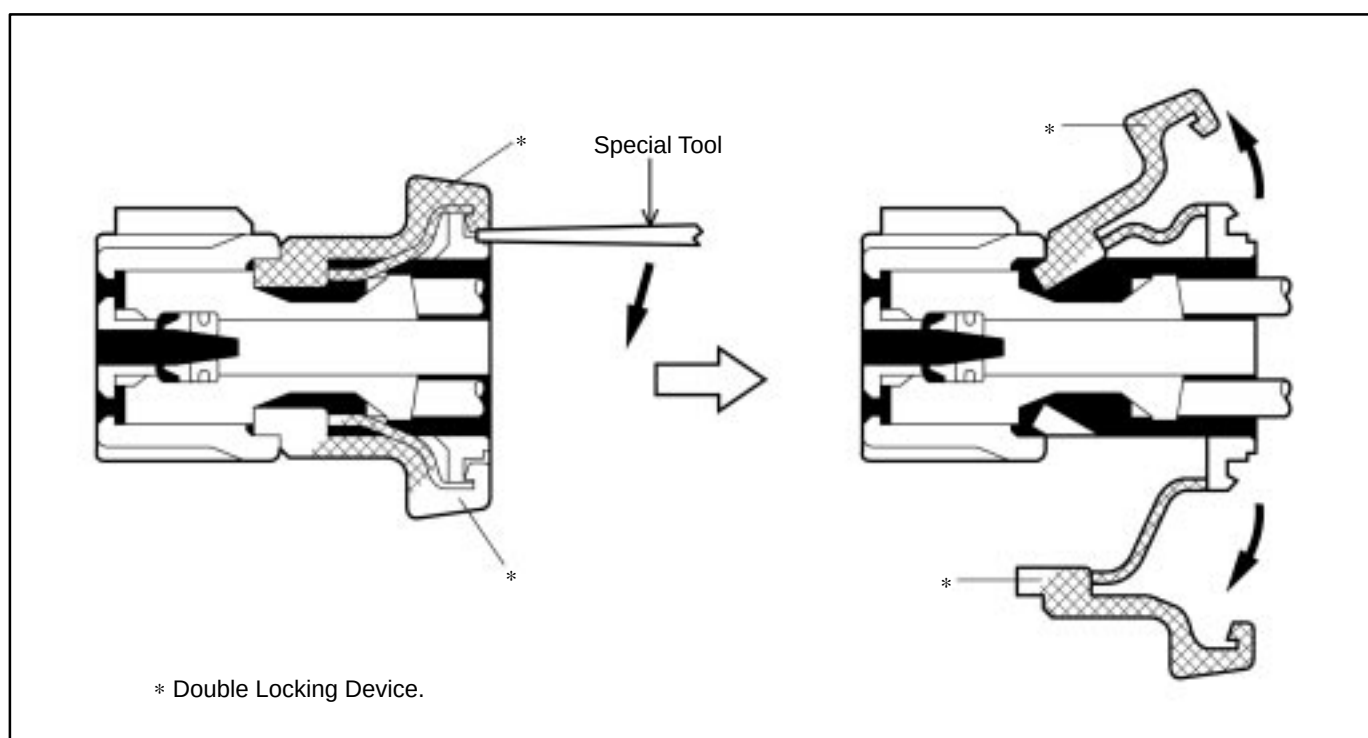
- Do not insert the special tool too much. It may cause damage on the fit section of the terminal and the wire harness.
- TNS housing is not reusable. Please replace it with the new one after replacement of the terminal.



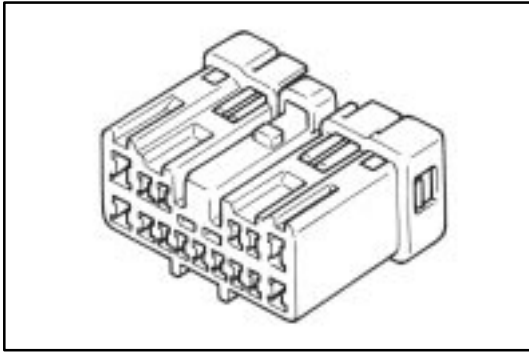
Type D (For 1.3 of Non-Waterproof Type Connector)

## [Case 1]

- (1) Use the special tool to unlock the double locking device.

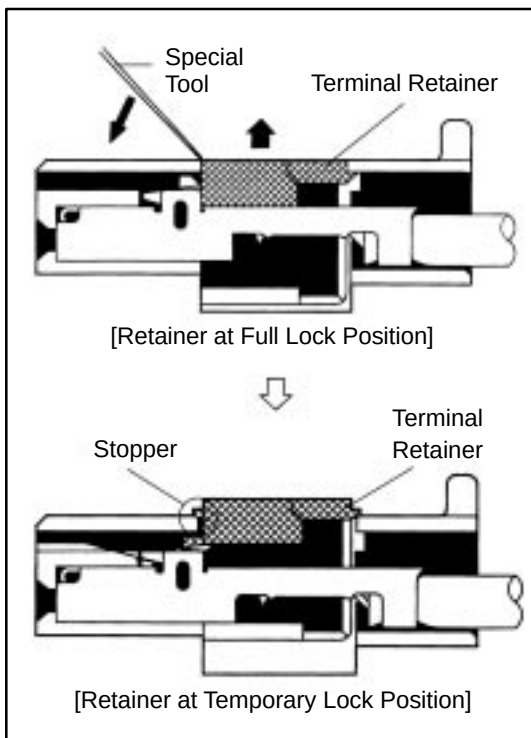
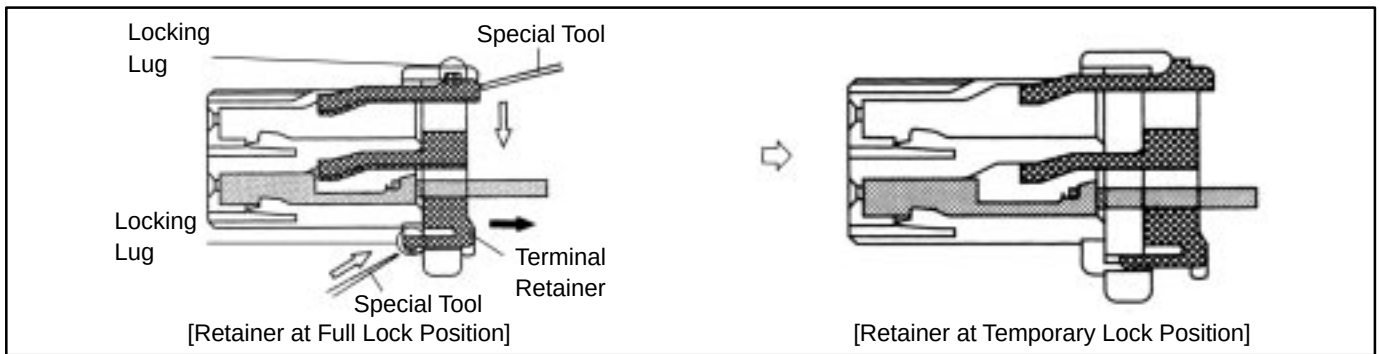


# TERMINAL AND CONNECTOR REPAIR-TERMINAL REPLACEMENT



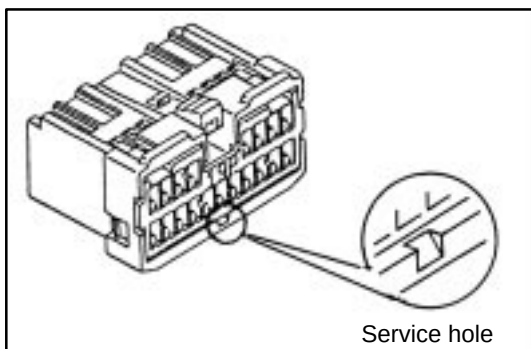
## [Case 2]

- (1) Using the special tool, push the terminal retainer locking lug (clip) and pull the terminal retainer up to the temporary lock position.



## [Case 3]

- (1) Using the special tool, raise the retainer up to the temporary lock position.



## [Case 4]